

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	A Scientometric Workbench to Guide Research & Innovation
<i>Company:</i>	UoN

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Data mining; Scientometrics

Project Background Information

Choosing what to research next involves understanding how a research field is changing, identifying connections with existing interests and deciding which directions deserve closer attention. Publication databases contain evidence that can inform these decisions, including changes in research topics, citation relationships and patterns of collaboration. In this project, your group will build a web-based system that helps researchers explore this evidence and discover potential directions for future publications.

The centrepiece will be a personalised research-field recommender. Users will describe their interests through keywords and selected papers, then receive suggestions based on topic relevance, publication trends and connections to related areas. Each recommendation will explain why it was selected and provide supporting publications. Users will be able to adjust their priorities, inspect how the recommendations change and save promising directions for further investigation. The system should make the assumptions and limitations of its recommendations understandable.

An interactive visual interface will allow users to explore the recommendations. An opportunity map could position topics according to

¹ An overview of different research areas in the school of computer science can be found [here](#).

relevance and recent growth, with publication volume represented by their size. Selecting a topic would reveal its development over time and the literature behind it. More ambitious extensions could reveal connections between research communities, the geography of collaboration and the citation paths linking earlier publications to current work. The aim is to produce a useful and visually engaging tool for investigating research opportunities.

Requirements

Core goals:

- A web application covering all core features.
 - Data collection and storage: retrieve metadata from an accessible source such as OpenAlex, cache it in a database and handle duplicate records and missing information.
 - Researcher profiles: allow users to describe their interests through keywords and selected papers (e.g., their own papers in the form of a BibTeX file).
 - Recommendation engine: rank research topics using relevance and publication trends, with adjustable weights for the recommendation criteria (supported by an appropriate UI).
 - Recommendation explanations: provide supporting publications, understandable explanations and information about data coverage.
 - Interactive visualisations: provide an opportunity map linked to topic timelines and publication details.
 - Saving/loading system for explorations: allow users to save recommendations.
- System evaluation: compare recommendations with popularity-only and relevance-only baselines and conduct a small task-based user evaluation.

In addition to those goals, there are a set of stretch goals for more advanced teams in order to produce a more compelling piece of software. Not all those stretch goals are necessary and there is an expectation that the final choice will depend on a discussion and negotiation between the project supervisor and the team.

Stretch goals:

- **Additional features**
 - A semantic research atlas: display individual papers in an interactive map with clustering, topic labels and navigation between related areas.
 - An animated collaboration map: show the geographic distribution of research activity and how institutional collaborations develop over time.
 - A citation explorer to provide expandable citation networks, temporal navigation and saved reading trails.
- **More advanced AI integration**

○ Self-improving recommendations: use researchers' relevance judgements to improve subsequent suggestions. ○ Richer recommendation methods: incorporate citation relationships and shared-reference similarity, with comparisons against the core approach. ○ Cross-field exploration: support comparisons and recommendations spanning multi and inter disciplinary research fields. ○ AI scientific advisor: Wrap all those data sources in a conversational AI assistant which can advise a researcher or a research team of the activity in their field. • Simple, quality-of-life features ○ Incremental updates: incorporate newly retrieved publications while preserving earlier datasets, configurations and results. ○ Collaborative investigation: allow users to share collections, annotate recommendations and assemble research reading lists. ○ Publication-quality exports: produce customisable figures and accompanying data suitable for reports and presentations. ○ Large-scale visualisation: maintain responsive interaction and stable, interpretable layouts as the publication collection grows.	
Skills	
<i>Highly desirable:</i> - Interest in NLP and data mining - Interest in web development - Interest in UX design	<i>Desirable:</i> - Experience in NLP and data mining - Experience in web development - Experience in UX design
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.